

Database

What is database?

- a database is an organized collection of data stored on a computer.
- managed by a Database Management System (DBMS).
- Data is stored in tables (rows and columns) so it's easy to access, manage and update.
- most databases use SQL to insert, update, delete, and read data.

Example:-

A school database may have a table students with columns like name, Roll No, class, marks.

Advantages of using Database.

1. NO duplicate data (minimizes redundancy).
2. Consistency - same data looks the same everywhere.
3. Data sharing - multiple user can access it.
4. Standard enforcement - follow the same rule.
5. Security - control who can see or change data.
6. integrity - ensures accuracy and completeness.

Type of Databases:-

1. Hierarchical Database
- Data stored in tree-like structure.
 - one parent can have many child records.
 - Example :- Organization chart.

2. Network Database:-

- Records can have many-to-many relationships.
- Example: A student can enroll in many courses; a course can have many students.

3. Relational Database (RDB)

- Data in tables linked by relationships (keys).
- Uses SQL for operations.
- Example:- MySQL, Oracle, PostgreSQL, MariaDB,

4. Non-Relational Database (NoSQL)

- data stored in non-tabular formats.
- Optimized for specific types of data
- Examples:- MongoDB, Redis, DynamoDB

RDBMS (Relational Database Management System) Key Concept

- Row $\rightarrow$ one record (e.g. a single student).
- Column $\rightarrow$ one attribute (e.g. Name, Age)
- Primary key $\rightarrow$ Unique ID for a record
- Foreign key $\rightarrow$ connects data between tables.

Types of NoSQL Database-

1. Document-oriented - stores data as documents (JSON/XML). Example: MongoDB.
2. Key-value store - simple key-value pairs, Example: Redis
3. Wide-column store - flexible columns per row. Example: Cassandra
4. Graph store - uses nodes and edges for relationships. Example: Neo4j.

SQL vs NOSQL

SQL

- structured data in tables
- Uses SQL queries
- But for complex queries

Example:- MySQL
Oracle

NOSQL

flexible, unstructured data

uses different query languages.

Best for large, varied, fast-changing data.

Examples:- MongoDB,
Redis